

Preeth Vijay

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[LinkedIn](#) | [Website](#) | [Github](#)

Education

University of Maryland, Global Campus

Major: BS in Computer Science **GPA:** 4.0

Graduating: October 2025

Certifications and Credentials

AWS Cloud Solutions Architect

Google UX Design Portfolio

Work Experience

DigitalT3 - Software Engineering Internship

APR 2023 - SEPT 2023

- **Product Design:** Drafted product specification and tech stack, found and navigated challenges.
- **Agile Backlog Management:** Maintained product backlog and tracked story points.
- **Data Extraction:** Used web scraping, regex, and OCR to extract and sanitize data with Python.
- **UX Prototyping:** Created wireframes and constructed the desired user flows.
- **Frontend Development:** Built polished UIs with HTML/CSS/JavaScript, Svelte, and Capacitor.
- **Backend Development:** Administered the AWS architecture. Wrote code to drive cloud systems.

Projects

macPB - Clipboard manager designed for macOS

CURRENT

Production-ready system utility that fills a market gap and will be available on the App Store!

- **Market Opportunity:** Identified that Windows / Linux have free clipboard managers, macOS doesn't.
- **Native Utility:** Closely integrated app with macOS to show my familiarity with Apple's ecosystem.
- **Rich Interface:** Utilized my UX design knowledge to deliver a friendly and robust user interface.

ADAM - Random number generator powered by chaos theory

MAY 2024

Bespoke PRNG written in pure C99 that produces ~6 GB of random data per second at 0.5 cycles/byte.

- **Statistically Sound:** Passed all known test suites, including NIST STS, DIEHARD, and TestU01.
- **Dual-Usage:** Implemented as both a library AND command line tool for macOS and Linux.
- **In-House Test Suite:** Included suite of 20+ statistical tests so users can easily analyze output quality.

LILITH - Implementation of a new cryptographic cipher

APR 2024

Cryptographic primitive written in Go which fully meets research specification and includes CLI.

- **Compliant:** Conformed to entire design in the paper *BLOSTREAM: A high speed stream cipher*.
- **Empirically Validated:** Passed the NIST Statistical Test Suite at multiple input sizes.
- **Resistant to Attacks:** Observed resistance to as many as 8 different classes of security attacks.

HammerPy - Auction art price estimation game

MAR 2024

Browse artwork from Artsy and Sotheby's, and see if you can guess the price!

- **Robust Main Menu:** Implemented controls to filter artwork scraping and even download full images.
- **Difficulty Levels:** Provided different confidence intervals to make guessing easier or harder.
- **Accessible Interface:** Designed simple GUI with tkinter featuring full keyboard navigation.

Skills

DESIGN: Figma, Photoshop, Illustrator, InDesign

WEB: HTML/CSS, JavaScript, SQL, AWS

SYSTEMS: C, Go, Swift, Python, Rust

MISC: Git, Unix, Bash, SIMD, UX Research